WORKED SOLUTION

# STEP I 2009, Question 2

Pure • Tangents to Cubic Curves

**STEP I 2009, Question 2**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q2

This question has a full guided treatment in *Beyond STEP*, Section 1.9, *Tangents to a Cubic*: https://beyondstep.keithmcnulty.org/ch01/tangents-cubic.html

## The key idea

A tangent meets a curve at its point of contact *twice*: when we substitute the tangent line into the curve's equation, the point of contact gives a repeated root. Once we know a repeated root of a cubic, the remaining root drops out by factorising, and that third intersection point turns out to have rational coordinates. Clearing denominators then turns the curve's equation into a sum of cubes.

## Solution

### The tangent at $(-a, b)$

First, $(-a, b)$ really is on the curve, since $x^3 + a^3 + b^3 = -a^3 + a^3 + b^3 = b^3 = y^3$.

Differentiating $y^3 = x^3 + a^3 + b^3$ implicitly with respect to $x$ (where $a$ and $b$ are constants) gives

$$3y^2 \frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2, \qquad \text{so} \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^2}{y^2}$$

wherever $y \neq 0$. At $(-a, b)$, where $b > 0$, the gradient is $a^2/b^2$. So the tangent is

$$\begin{aligned} y - b &= \frac{a^2}{b^2}(x + a) \\ b^2 y - b^3 &= a^2 x + a^3 \\ b^2 y - a^2 x &= a^3 + b^3, \end{aligned}$$

as required.

**The intersections when $a = 1, b = 2$**

Now the curve is $y^3 = x^3 + 9$ and the tangent at $(-1, 2)$ is $4y - x = 9$, that is, $y = \frac{1}{4}(x + 9)$. Substituting into the curve's equation:

$$\begin{aligned} \frac{(x+9)^3}{64} &= x^3 + 9 \\ x^3 + 27x^2 + 243x + 729 &= 64x^3 + 576 \\ 0 &= 63x^3 - 27x^2 - 243x - 153. \end{aligned}$$

Every coefficient is divisible by 9, and dividing through gives

$$7x^3 - 3x^2 - 27x - 17 = 0,$$

which is the cubic we were asked for.

**Finding $p, q, r, s$**

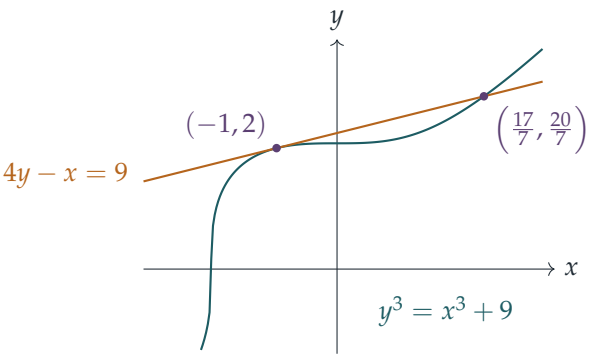
**Ask yourself** We already know one root of this cubic. Why should it be a *double* root?

The line touches the curve at $x = -1$, so $x = -1$ is a repeated root: near the point of contact the line and the curve agree to first order, so their difference vanishes to second order there. Hence $(x + 1)^2 = x^2 + 2x + 1$ is a factor. Comparing the $x^3$ and constant terms, the other factor must be $7x - 17$, and we can check this:

$$(x + 1)^2(7x - 17) = 7x^3 + 14x^2 + 7x - 17x^2 - 34x - 17 = 7x^3 - 3x^2 - 27x - 17.$$

So the tangent meets the curve again where $x = \frac{17}{7}$, and there

$$y = \tfrac{1}{4}\left(\tfrac{17}{7} + 9\right) = \tfrac{1}{4} \cdot \tfrac{80}{7} = \tfrac{20}{7}.$$




Since this point lies on $y^3 = x^3 + 9$,

$$\left(\frac{20}{7}\right)^3 = \left(\frac{17}{7}\right)^3 + 9.$$

Multiplying by $7^3$ gives $20^3 = 17^3 + 9 \cdot 7^3$. The last term is not a cube, but $9 = 1 + 8 = 1^3 + 2^3$, so

$$9 \cdot 7^3 = 7^3 + 2^3 \cdot 7^3 = 7^3 + 14^3.$$

(This is where the original form $a^3 + b^3$ with $a = 1$, $b = 2$ comes back.) Therefore

$$20^3 = 17^3 + 14^3 + 7^3,$$

and we can take $p = 20$, $q = 17$, $r = 14$, $s = 7$ (the last three in any order).

## Checks

Directly: $17^3 + 14^3 + 7^3 = 4913 + 2744 + 343 = 8000 = 20^3$. Also, the three roots of the cubic are $-1, -1, \frac{17}{7}$, whose sum is $\frac{3}{7}$ and product is $\frac{17}{7}$, which matches $3/7$ and $17/7$ read off from the coefficients. A numerical derivative of $y = \sqrt[3]{x^3 + a^3 + b^3}$ at $x = -a$ agrees with $a^2/b^2$.

## Common slips

- Not using the double root: trying to solve the cubic from scratch instead of dividing out $(x + 1)^2$.
- Stopping at $20^3 = 17^3 + 3087$. We need positive integers whose cubes add up, and $3087 = 9 \cdot 7^3$ has to be split as $7^3 + 14^3$.

## Beyond STEP

This is the classical *tangent method* for curves of degree three. A line meets a cubic curve in three points, counted with multiplicity, so two rational points (or one counted twice, as here) produce a third. On elliptic curves this chord-and-tangent construction becomes a group law, the starting point of a large area of modern number theory. Davenport's *The Higher Arithmetic* discusses Diophantine equations of this kind, and Silverman and Tate's *Rational Points on Elliptic Curves* develops the geometry in full.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.